This is one of the simplest, and yet most used data structures. It’s provided natively by most programming languages. An *array* is a collection of homogeneous data. At a low level it is, roughly, a chunk of memory where elements are stored contiguously. Usually programming languages only provide *static arrays*: their size can't change and the number of elements they store needs to be decided when they are created (or at most at initialization). *Dynamic arrays*, however, can grow when new elements are added and shrink when elements are removed: figure C.1 provides an example of how this works. It's not too hard to find *dynamic arrays* in mainstream languages: *JavaScript* arrays are intrinsically dynamic.

It can be proven that it's possible to implement dynamic arrays in such a way that, *on aggregate*[1], insertion and removal are as fast as for static arrays.

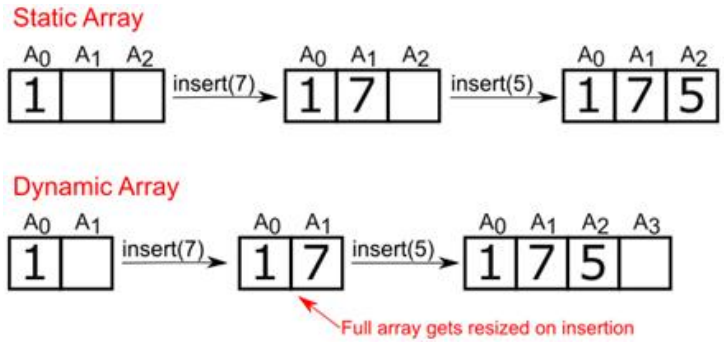


Figure C.1 An example of insertion into static (top) and dynamic arrays (bottom). Notice how the size of the static array in the upper half is constant (always have 3 elements), while the dynamic array in the other half starts with size 2, and only when an element is added to a full array, its size is increased (doubled, actually). In this example, the static array could not have any other element inserted, unless we override another one.

Elements in arrays must all have the same type and require the same space to be stored[2]. When this specification is met, it makes easier to